



Small Intestine — MCQ Practice

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. A radiologist describes the first and shortest region of the small intestine, about 25 cm long, curving around the head of the pancreas. Which region is this?

EASY

- A. Pyloric sphincter
- B. Ileum
- C. Ileocecal valve
- D. Duodenum
- E. Jejunum

2. A pathologist finds lymphoid aggregates containing B and T lymphocytes that give immune surveillance in the small intestine. Where are they found?

EASY

- A. Jejunum
- B. Serosa of the duodenum
- C. Duodenal submucosa
- D. Pyloric sphincter
- E. Ileum (Peyer's patches)

3. Which motility pattern involves local contractions that mix chyme with digestive secretions?

EASY

- A. Micelle formation
- B. Emulsification
- C. Segmentation
- D. Peristalsis
- E. Re-esterification

4. Roughly what proportion of the nutrients and water that pass through the digestive system is absorbed by the small intestine?

EASY

- A. About 90%
- B. About 50%
- C. About 10%
- D. About 25%
- E. About 70%

5. A patient is given a drug that mimics sympathetic stimulation of the small intestine. Which effect is expected?

MEDIUM

- A. Decreased motility with increased blood flow
- B. Decreased motility and decreased blood flow
- C. Increased motility with decreased secretion
- D. Loss of segmentation only, with normal peristalsis
- E. Increased peristalsis and increased secretion

6. Which enzymes begin carbohydrate digestion, and what do they produce from polysaccharides?

MEDIUM

- A. Pepsin, producing disaccharides
- B. Salivary amylase only, producing monosaccharides
- C. Salivary and pancreatic amylase, producing disaccharides
- D. Pancreatic lipase, producing monoglycerides
- E. Brush border sucrase and maltase, producing monosaccharides

7. A patient has normal amylase but lacks the brush border enzymes lactase, maltase and sucrase. Which carbohydrate products would fail to be absorbed?

MEDIUM

- A. Fructose, because it needs sucrase to be transported
- B. Disaccharides, because only monosaccharides are absorbed
- C. Glucose, because it needs lactase to cross the membrane
- D. Galactose, because it requires bile salts
- E. Polysaccharides, which are normally absorbed directly

8. Dipeptides and tripeptides cross the apical membrane of the enterocyte. Which statement is correct?

MEDIUM

- A. They use Na⁺-dependent amino acid transporters and stay intact in the cell
- B. They enter the lacteals as chylomicrons
- C. They travel in micelles and are re-esterified
- D. They are hydrolysed by pepsin at the brush border before uptake
- E. They use the proton-coupled transporter PepT1 and are hydrolysed to amino acids inside the cell

9. A patient with a bile duct obstruction cannot release bile into the duodenum. Which step of fat handling is most directly impaired?

MEDIUM

- A. Emulsification of lipids into micelles
- B. Absorption of water-soluble vitamins by diffusion
- C. Secretion of intestinal juice by mucosal glands
- D. Hydrolysis of disaccharides at the brush border
- E. Na⁺-dependent uptake of amino acids

10. A patient with severe lipid malabsorption is at risk of deficiency of which group of vitamins?

MEDIUM

- A. A, D, E and K
- B. Water-soluble vitamins in general
- C. B vitamins only
- D. Vitamin B12 only
- E. Vitamins B and C

11. Which statement about calcium absorption is correct?

MEDIUM

- A. It occurs in the ileum and needs intrinsic factor
- B. It occurs in the duodenum and is vitamin D-dependent
- C. It is enhanced by gastric acid but not by vitamin D
- D. It occurs mainly in the jejunum with bile salts
- E. It occurs throughout the small intestine independently of vitamin D

12. A patient with markedly reduced gastric acid develops iron deficiency. Which statement best explains this?

MEDIUM

- A. Iron absorption is coupled with Na⁺ and glucose
- B. Iron is absorbed via micelles with dietary lipids
- C. Iron absorption depends on vitamin D in the jejunum
- D. Iron absorption is enhanced by gastric acid and occurs mainly in the duodenum
- E. Iron is absorbed in the ileum together with intrinsic factor

13. A pathologist notes prominent circular folds (plicae circulares), abundant villi and a rich blood supply in a segment of small intestine. Which region is this?

MEDIUM

- A. Pyloric sphincter
- B. Ileocecal valve
- C. Duodenum
- D. Ileum
- E. Jejunum

14. Inside enterocytes, absorbed lipid products are re-esterified into triglycerides. What do these triglycerides combine with to form chylomicrons?

MEDIUM

- A. Monosaccharides and fructose
- B. Bile salts and micelle proteins
- C. Vitamin B12 and intrinsic factor
- D. Cholesterol, phospholipids and proteins
- E. Amino acids and small peptides

15. A patient has a normal ileum but no intrinsic factor. Which outcome best fits the lecture?

HARD

- A. Impaired vitamin B12 absorption, because B12 needs intrinsic factor for absorption in the ileum
- B. Impaired fat-soluble vitamin absorption, because intrinsic factor forms micelles
- C. Normal B12 absorption, because the ileum takes up B12 regardless of intrinsic factor
- D. Impaired bile salt uptake, because intrinsic factor is needed in the jejunum
- E. Impaired calcium absorption, because intrinsic factor is vitamin D-dependent

16. An inhibitor blocks the sodium-dependent glucose co-transporter in the enterocyte. Which absorption is most affected?

HARD

- A. Glucose only, since galactose uses facilitated diffusion
- B. Glucose and galactose, while fructose is still absorbed by facilitated diffusion
- C. Fructose only, because it depends on Na^+ co-transport
- D. Amino acids, because they share the same carrier
- E. All monosaccharides equally, because all depend on Na^+

17. A patient's thoracic duct is obstructed. Which absorbed-nutrient route is most compromised?

HARD

- A. Na^+ -coupled uptake of glucose across the enterocyte
- B. Portal transport of amino acids to the liver
- C. Transport of triglycerides in chylomicrons via the lacteals and lymph
- D. Uptake of bile salts in the ileum
- E. Portal transport of monosaccharides to the liver

18. A patient has impaired segmentation but intact peristalsis. Which consequence is most likely?

HARD

- A. Chylomicron formation fails inside the enterocytes
- B. Chyme stops moving altogether, because peristalsis needs segmentation
- C. Mixing with secretions increases, improving absorption
- D. Brush border enzymes are lost, because they depend on segmentation
- E. Chyme still moves forward but mixes less with secretions and contacts the mucosa less, reducing digestion and absorption

19. Which sequence correctly describes protein digestion and absorption?

HARD

- A. Amylase, then brush border sucrase, then the sodium-glucose co-transporter into the blood
- B. Pepsin, then bile salts and micelles; amino acids diffuse in and enter the lacteals
- C. Pepsin (stomach), then pancreatic trypsin, chymotrypsin and carboxypeptidase, then brush border peptidases; amino acids and small peptides are absorbed and go to the liver
- D. Pancreatic lipase, then brush border peptidases, then micelles and lacteals into the thoracic duct
- E. Brush border peptidases, then pepsin, then PepT1 and chylomicron formation in the lymph

20. Which statement about electrolyte absorption in the small intestine is correct?

HARD

- A. Na^+ enters only through Na^+ channels; Cl^- is actively pumped in; K^+ is co-transported with glucose
- B. Na^+ enters only with nutrients; Cl^- is exchanged with K^+ ; K^+ uses facilitated diffusion through PepT1
- C. Na^+ absorption is independent of nutrients; Cl^- moves in micelles; K^+ leaves through the lacteals
- D. Na^+ is coupled with nutrient absorption and also enters via Na^+ channels; Cl^- is passive or exchanged with HCO_3^- ; K^+ moves mainly by diffusion
- E. Na^+ enters by diffusion alone; Cl^- is co-transported with amino acids; K^+ is actively secreted

✓ Answer key

Q	Answer	Level	Why (lecture slide)
1	D	EASY	The duodenum is the first and shortest region (~25 cm) and curves around the head of the pancreas. [Slide 12]
2	E	EASY	Peyer's patches are lymphoid aggregates in the ileum with B and T lymphocytes. [Slide 18]
3	C	EASY	Segmentation = local contractions mixing chyme with secretions; peristalsis moves chyme forward. [Slide 21]
4	A	EASY	The small intestine absorbs ~90% of nutrients and water. [Slide 37]
5	B	MEDIUM	Sympathetic stimulation ↓ ↓ motility and blood flow; parasympathetic ↑ ↑ peristalsis and secretion. [Slide 35]
6	C	MEDIUM	Digestion begins with salivary and pancreatic amylase (polysaccharides → disaccharides); brush border enzymes then give monosaccharides. [Slide 22]
7	B	MEDIUM	Brush border enzymes convert disaccharides into monosaccharides; only monosaccharides (glucose, galactose, fructose) are absorbed. [Slides 20, 22]
8	E	MEDIUM	Di- and tripeptides use PepT1 (proton-coupled) and are hydrolysed into amino acids inside the enterocyte. [Slide 26]
9	A	MEDIUM	Lipids require emulsification by bile salts into micelles; micelles carry lipid products to the epithelial surface. [Slide 28]
10	A	MEDIUM	Fat-soluble vitamins (A, D, E, K) are absorbed with dietary lipids via micelles. [Slide 30]
11	B	MEDIUM	Calcium absorption occurs in the duodenum and is vitamin D-dependent. [Slide 31]
12	D	MEDIUM	Iron absorption is enhanced by gastric acid and occurs mainly in the duodenum. [Slide 31]
13	E	MEDIUM	The jejunum is richly supplied with blood vessels, has prominent circular folds and abundant villi for absorption. [Slides 14, 15]
14	D	MEDIUM	Triglycerides combine with cholesterol, phospholipids and proteins to form chylomicrons, which enter lacteals. [Slide 29]
15	A	HARD	The ileum is specialised for B12 uptake, but B12 requires intrinsic factor for absorption in the ileum. [Slides 17, 30]
16	B	HARD	Glucose and galactose use sodium-dependent co-transport; fructose uses facilitated diffusion. [Slides 23, 24]
17	C	HARD	Chylomicrons enter lacteals and travel by the lymphatic system (thoracic duct); sugars and amino acids enter blood → liver. [Slides 29, 34]
18	E	HARD	Segmentation mixes chyme with digestive juices and brings food into contact with the mucosa; peristalsis propels chyme. [Slides 21, 37]
19	C	HARD	Pepsin → pancreatic enzymes → brush border peptidases → AAs/small peptides absorbed → blood → liver. [Slides 25, 26]
20	D	HARD	Na ⁺ : coupled with nutrients and via Na ⁺ channels; Cl ⁻ : passive or exchange with HCO ₃ ⁻ ; K ⁺ : mainly diffusion. [Slide 32]